Recitation 8: Solow Growth Model

Consider an economy described by the Solow growth model where the production function is given by

$$Y = K^{1/2}L^{1/2},$$

where K is the capital stock and L is labor, which remains constant over time. Assume that the savings rate (s) is 0.25, the depreciation rate (δ) is 0.05, and the labor force (L) is normalized to 1.

1. Calculate the steady-state level of capital (K^*) and output (Y^*).
2. Suppose the savings rate increases to 0.35. What will be the new steady-state level of capital and output?
3. Discuss the transition dynamics. How does the economy transition from the old steady-state to the new one after the increase in the savings rate?

Solution:

1. Original Steady-State Calculation

In the steady-state, $K_{t+1} = K_t = K^*$. The steady-state condition for capital can be derived as follows:

$$0 = s \cdot Y^* - \delta \cdot K^*.$$

Given the production function $Y = K^{1/2}$ (since $L = 1$), the steady-state condition for capital is:

$$0 = s\sqrt{K^*} - \delta \cdot K^*.$$

Here, $s = 0.25$ and $\delta = 0.05$. We rearrange the steady-state condition to find K^*:

$$0 = 0.25\sqrt{K^*} - 0.05 \cdot K^*$$

$$0.05 \cdot K^* = 0.25\sqrt{K^*}$$

$$K^* = 5\sqrt{K^*}$$

$$(K^*)^2 = 25 \cdot K^*.$$

Given that $K^* \neq 0$, we divide both sides by K^* to get:

$$K^* = 25.$$

Now, to find the steady-state output Y^*:

$$Y^* = \sqrt{K^*} = \sqrt{25} = 5.$$

2. New Steady-State with Increased Savings Rate

Now let's consider the scenario where the savings rate increases to $s = 0.35$. We will find the new steady-state levels of capital (K^{**}) and output (Y^{**}):

$$0 = 0.35\sqrt{K^{**}} - 0.05 \cdot K^{**}$$

$$0.05 \cdot K^{**} = 0.35\sqrt{K^{**}}$$
$$K^{**} = 7\sqrt{K^{**}}$$
$$(K^{**})^2 = 49 \cdot K^{**}.$$

So K^{**} would be:

$$K^{**} = 49.$$

And the new steady-state output Y^{**} is:

$$Y^{**} = \sqrt{K^{**}} = \sqrt{49} = 7.$$

3. Transition Dynamics

When the savings rate increases from 0.25 to 0.35, the economy doesn't instantly jump to the new steady state. Instead, the higher savings rate leads to faster accumulation of capital:

- Before the change, the economy was adding $0.25 \times Y_t - 0.05 \times K_t$ to the capital stock each period.
- After the increase, the economy adds $0.35 \times Y_t - 0.05 \times K_t$ to the capital stock each period.

This increased investment raises the capital stock over time, driving up output due to the production function until the new steady-state capital stock of 49 and output of 7 are reached. During this transition, the economy experiences growth in per capita output until the steady-state is reached, at which point growth in per capita output ceases again (in the absence of technological progress or further changes in the savings rate).